CHAPTER 7

Environment

7.1. Space story

The Pioneer 10 and 11 spacecraft were the first deep-space missions. The spacecraft were launched in 1972 and 1973 and were the first to pass through the asteroid belt. In the 1980s it was observed that both spacecraft were slowing. Some scientists speculated that Einstein's General Theory of Relativity was incorrect. The problem was eventually traced by Slava Turyshev of JPL to thermal radiation from the spacecraft. The heat, like any radiation, produced a thrust decelerating the two probes [1].

7.2. Introduction

The space environment is all of the external influences on the spacecraft [2]. The environment is the source of force and torque disturbances on the spacecraft. It also is a source of radiation that damages electronics and plasma that can charge the spacecraft. The latter effects do not have a direct influence on spacecraft control but indirectly influence the control-system design as they drive requirements for more robust electronics.

This chapter covers the space environment. The disturbance effects of the space environment are covered in the disturbances chapter.

7.3. Optical environment

The optical environment includes all electromagnetic frequencies from infrared wavelengths to visible bands. Sources of optical influx are the Sun, planets, and other spacecraft. X-rays and gamma-rays are considered ionizing radiation sources.

7.3.1 Solar radiation

Solar pressure is the dominant disturbance on a spacecraft in geosynchronous and interplanetary orbit. It is also used for propulsion by solar sails. Solar pressure is due to the force of photons on the surfaces of the spacecraft. The pressure is

$$s = \frac{p}{c} \tag{7.1}$$

where p is the power of the sunlight and c is the speed of light. As can be seen, a 1-N force would require 3×10^8 W. Thus a flashlight-propelled spacecraft would take a long

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time to gain any velocity. The solar flux is

$$p = \frac{1367}{\left(\frac{r}{r_e}\right)^2} \tag{7.2}$$

where r_e is the mean distance of the Earth from the Sun. The units are W/m^2.

7.3.2 Earth albedo

Albedo is the reflection of sunlight off of the surface and clouds of the Earth. As the Earth cannot be treated as a point source (as can the Sun) and the reflectivity varies over the surface, albedo is difficult to model accurately [3]. The simplest model is to assume that reflection from the Earth's surface and clouds is diffuse [4]. The flux from the total reflectance can then be computed by integrating it over the surface of the Earth.

The Earth's surface is modeled as a Lambertian reflector as shown in Fig. 7.1.

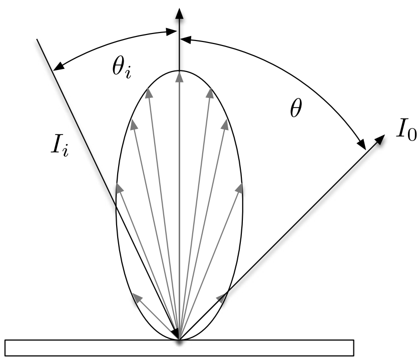


Figure 7.1 Lambertian reflector.

If the spacecraft is more than five times the distance from the surface as the dimensions of the surface, then the following equation can be used to model the flux [5]:

$$\Phi_{ij} = \alpha PA \frac{\cos\theta_i \cos\theta_j}{\pi r^2} \tag{7.3}$$

where α is the absorption coefficient for the thermal flux, P is the solar flux, A is the area of the surface patch, $\cos\theta_i$ is the angle between the normal of the surface patch and the incoming flux, and $\cos\theta_j$ is the angle between the normal of the spacecraft patch and the incoming radiation. For a spacecraft in a 200-km orbit, this would mean that the triangles representing the surface of the Earth could be no more than 40 km across.

This requires an active subdivision of the Earth's surface so that only the triangles under the spacecraft are tessellated. The α coefficient varies with each patch and may represent ice, water, vegetation, clouds, etc. The spot under the spacecraft was subdivided into smaller triangles. The size of the triangles decreases with angular distance from the spacecraft vector.

7.3.3 Earth radiation

Earth radiation is assumed uniform over the surface of the Earth. The flux is

$$q = \frac{400}{\left(\frac{r}{r_E}\right)^2} \tag{7.4}$$

7.4. Atmosphere

The atmosphere interacts with the spacecraft producing disturbance forces and torques. In low-Earth orbit, drag compensation may be necessary to prevent premature reentry. It may also be necessary to maintain the desired phasing in a constellation.

Many models are used for atmospheric density. The simplest is an exponential atmosphere. This covers the entire atmosphere with a single equation but cannot fit atmospheric-density variations accurately. For the Earth, the exponential atmosphere model is

$$\rho = 1.225 e^{-0.0817 h^{1.15}} \tag{7.5}$$

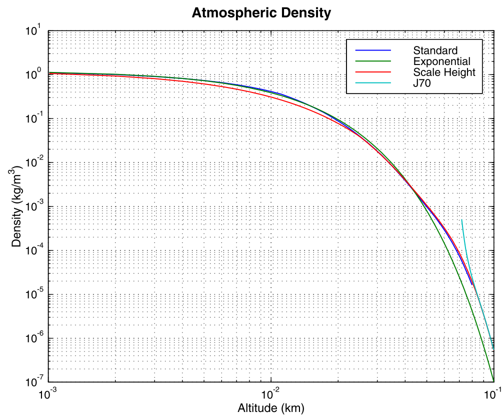
where h is the altitude in km. This is a variation on a true exponential model that is

$$\rho = \rho_0 e^{-\frac{h}{h_s}} \tag{7.6}$$

where h_s is the scale height and ρ_0 is the surface density. This can be expanded to a scale-height model. This is an exponential model that fits the model at many altitudes. However, the atmosphere does not follow an exponential law in all ranges of altitudes. The Sun has a major influence on the atmosphere and the density seen by the spacecraft is a function of the angle between the Sun and the spacecraft. A model that includes this effect is the Jacchia atmosphere model but it is only valid for altitudes > 70 km. The last is the standard atmosphere that is only valid under 18.6 km. The last curve fits actual data and is the best for atmosphere studies. The Jacchia model includes the effect of the Sun that is important for low-Earth orbiting satellites. The models are compared in Example 7.1.

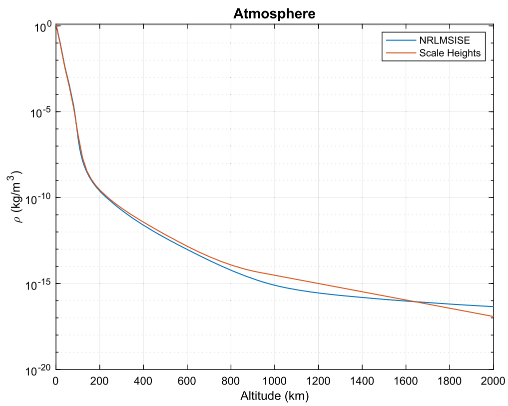
Other models exist including the MSIS model [6]. The NRLMSISE-00 model was developed by Mike Picone, Alan Hedin, and Doug Drob. It is based on extensive data from spacecraft. Example 7.2 compares it with scale heights. The results are close but scale heights do not include the Sun effects that can be important.

```
altitude = linspace(0,100);
z = StdAtm( altitude*1000 ); % standard
d = zeros(4,100);
k = 1:length(z.density);
d(1,k) = z.density;
% exponential
d(2,:) = AtmDens1(altitude);
% scale heights
d(3,:) = AtmDens2(altitude);
p = struct('aP', 400, 'dd', 79, ...
           'f', 230, 'fHat', 230, ...
           'fHat400',230,'lat',-30,...
           'lng', 0, 'mm', 840,...
           'yr', 1970 );
j = find( altitude >= 71 );
for k = j   % Jacchia 1970
  p.z = altitude(k);
  [rho, nHe, nN2, nO2, nO, tZ, eM] ...
      = AtmJ70( p );
  d(4,k) = rho*1000;
end
Plot2D( altitude/1000, d, ...
  'Altitude␣(km)','Density␣(kg/m^3)',...
  'Atmospheric␣Density', 'ylog')
legend('Standard', 'Exponential', ...
       'Scale␣Height', 'J70' )
```



Example 7.1: Atmosphere-density models.

```

%% NRL Model

for i = 1:8
      aph.a(i)=100;
end
input(1) = struct('year',0,'doy',172,'sec',29000,
      'alt',0,'lst',0,'g_lat',60,'g_long'
      ,-70,...
  'f107A',150,'f107',150,'ap',4,'ap_a',aph);


z     = zeros(1,24);
flags = struct('switches',z,'sw',z,'swc',z);

flags.switches(1) = 0;
flags.switches(10)=1; % Does not use daily ap

for i = 2:24
      flags.switches(i)=1;
end

alt = linspace(0,2000);
rho = zeros(1,length(alt));
input(1).ap_a = aph;
for k = 1:length(alt)
  input(1).alt = alt(k);
  flags.switches(1) = 0;
      output = AtmNRLMSISE( 'gtd7', [], input
         (1), flags );
  rho(k) = output.d(6);
end

rho2 = AtmDens2( alt );

Plot2D(alt,[rho*1000;rho2],'Altitude␣(km)','\rho␣
      (kg/m^3)','Atmosphere','ylog')

legend('NRLMSISE','Scale␣Heights')
```



Example 7.2: Comparison of NRLMSISE with scale heights.

The most accurate model is the Air Force "High Accuracy Satellite Drag Model" (HASDM) [7]. This model uses satellite orbit data to refine the model. It is a great improvement over the models cited above. As Bowman states [8]

> *Current thermospheric density models do not adequately account for dynamic changes in atmospheric drag for orbit predictions, and no significant operational improvements have been made since 1970. Lack of progress is largely due to poor model inputs in the form of crude heating indices, as well as poor model accuracy and resolution, both spatial and temporal.*

The European Space Agency has a similar initiative that might also be a source of improved models [9].

For planetary missions, the atmospheres of the target planets must be known. Densities for Titan and Neptune are shown in Fig. 7.2.

7.5. Plasma

A plasma is a gas in which the atoms are stripped of their electrons to that they have a net positive charge. The gas comprises both the ions and the electrons so that globally it is neutral. A spacecraft will collect charge and gain an electrostatic potential [10]. Charging can lead to surface arcing that can destroy electronics. For this reason, attitude control electronics should be located so arcs do not touch the electronics. This can be a problem for sensors that need to be located on the surface of the spacecraft. Design techniques to minimize differential charging are well understood and need to be employed [13], [11].

The attitude control system should have procedures in place in case a major component is lost due to charging. For example, a ground control loop can substitute for a damaged reaction wheel or flight computer.

7.6. Gravity

Gravity influences the orbit of the spacecraft and creates the gravity-gradient torque on the spacecraft. The following sections discuss gravity models. Gravity gradient is discussed in the chapter on disturbances.

7.6.1 Point mass

The point-mass model assumes that the mass is concentrated at an infinitesimal point at the center of the planet. The gravitational acceleration is

$$a = -\mu \frac{r}{|r|^3} \tag{7.7}$$

where μ is the gravitational parameter and r is the vector from the point mass to the spacecraft.

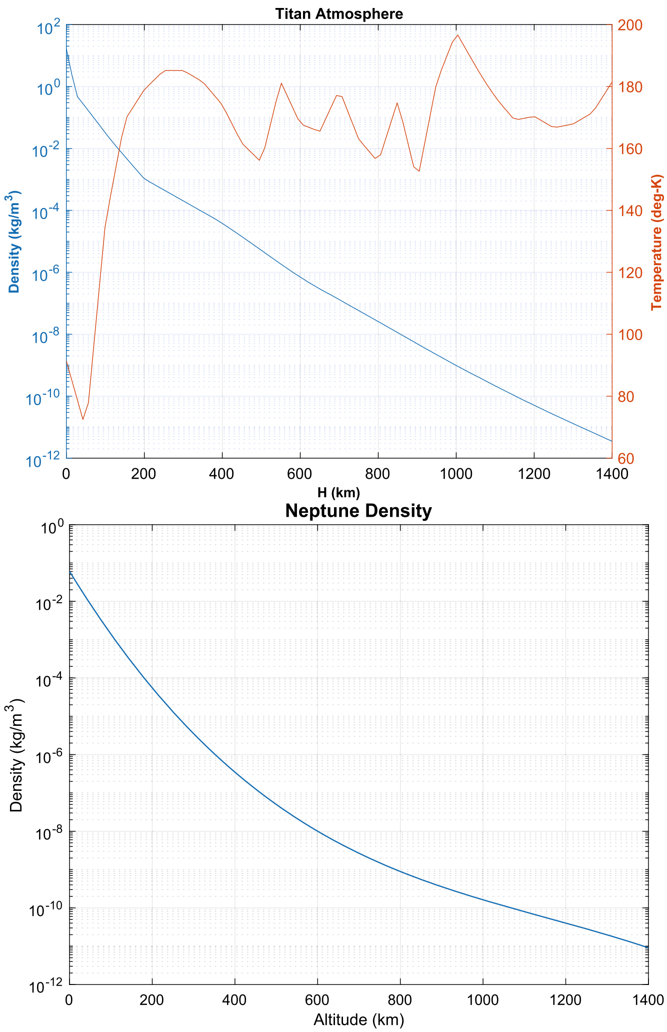


Figure 7.2 Atmospheric properties for Titan and Neptune.

7.6.2 Spherical harmonics

The spherical-harmonic expansion method works as follows. Define the perturbing gravitational potential of a planet as

$$V = \frac{\mu}{r} \sum_{n=2}^{\infty} \left[\left(\frac{a}{r}\right)^n \sum_{m=0}^{\infty} \left(S_{n,m} \sin m\lambda + C_{n,m} \cos m\lambda\right) P_{n,m}(\sin\phi) \right] \tag{7.8}$$

defined in the planet fixed frame. a is the radius of the planet. The definition of the coordinates is shown in Fig. 7.3.

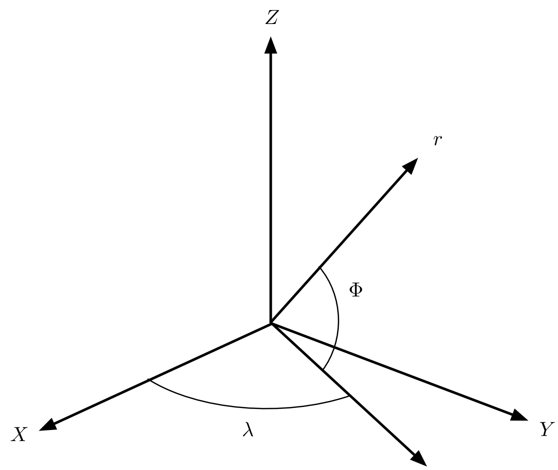


Figure 7.3 Coordinates for a nonhomogeneous planet.

If we define i, j, k as unit vectors along the planetary x–, y–, and z–axes, the gravitational acceleration can be found as follows

$$\frac{\partial V}{\partial r} = \sum_{n=2}^{\infty}\sum_{m=0}^{n}\frac{\partial V_{n,m}}{\partial r} \tag{7.9}$$

and

$$\frac{\partial V_{n,m}}{\partial r} = \frac{\mu}{r^2}\left(\frac{a}{r}\right)^n\left[-r(vH_{n,m} + B_{n,m}) + iD_{n,m} - jE_{n,m} + kH_{n,m}\right] \tag{7.10}$$

$$B_{n,m} = (C_{n,m}\hat{C}_m + S_{n,m}\hat{S}_m)(n + m + 1)P_n^m \tag{7.11}$$

$$\begin{aligned}
E_{n,m} &= -m(C_{n,m}\hat{S}_{m-1} - S_{n,m}\hat{C}_{m-1})\\
D_{n,m} &= m(C_{n,m}\hat{C}_{m-1} + S_{n,m}\hat{S}_{m-1})\\
H_{n,m} &= (C_{n,m}\hat{C}_m + S_{n,m}\hat{S}_m)P_n^{m+1}\\
\hat{C}_m &= \hat{C}_1\hat{C}_{m-1} - \hat{S}_1\hat{S}_{m-1}\\
\hat{S}_m &= \hat{S}_1\hat{C}_{m-1} + \hat{C}_1\hat{S}_{m-1}
\end{aligned}$$

The starting conditions are

$$\begin{matrix} \hat{C}_0 = 1 & \hat{S}_0 = 0 \\ \hat{C}_1 = \frac{x}{r} & \hat{S}_1 = \frac{y}{r} \end{matrix} \tag{7.12}$$

The derivatives of the Legendre polynomials are

$$P_n^0 = \frac{1}{n}\left[(2n-1)\nu P_{n-1}^0 - (n-1)P_{n-2}^0\right] \tag{7.13}$$

$$P_n^m = P_{n-2}^m + (2n-1)P_{n-1}^{m-1} \tag{7.14}$$

$$P_0^0 = 1$$

$$P_1^0 = \frac{z}{r} \equiv \nu$$

$$P_0^1 = 0$$

$$P_1^1 = 1$$

This kind of harmonic expansion is the standard method for modeling a planet's gravitational field. Harmonic models exist for the Earth, Moon, Mars, and other planets. A harmonic model is convenient because it gives an idea of how much influence higher-order terms have on the overall force.

An Earth 68-by-68 spherical-harmonic gravity model is shown in Fig. 7.4.

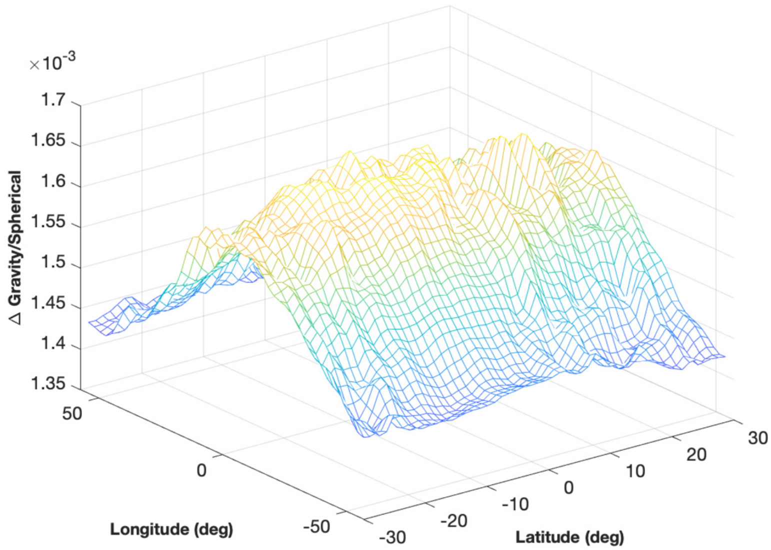


Figure 7.4 Earth 68-by-68 spherical-harmonic gravity model.

The lunar 150-by-150 spherical-harmonic gravity model is shown in Fig. 7.5. Note the "lumps" that are indicative of mass concentrations.

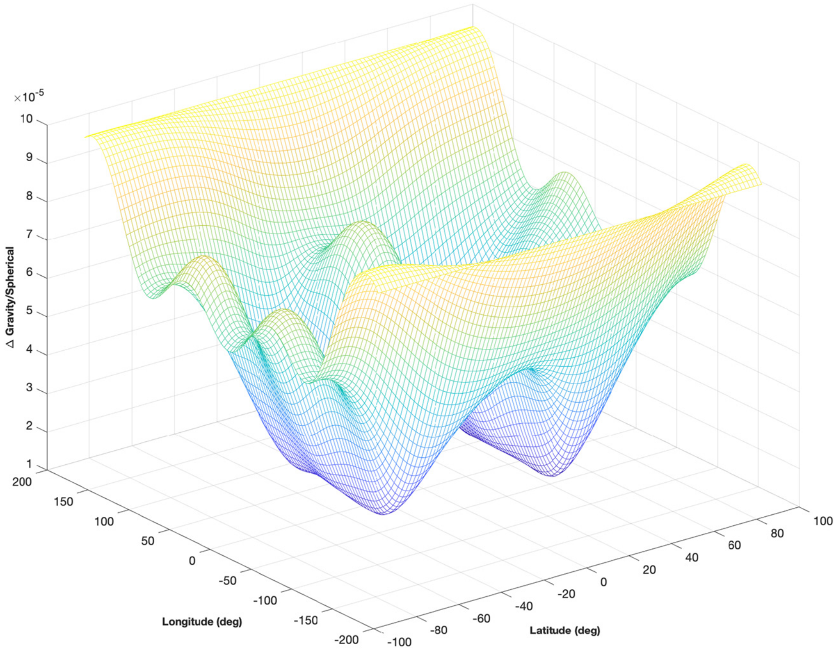


Figure 7.5 150-by-150 spherical-harmonic lunar gravity model. The plot shows magnitude variations.

7.7. Magnetic fields

The following planets and moons have significant magnetic fields

1. Earth;
2. Mars;
3. Jupiter;
4. Saturn;
5. Uranus;
6. Neptune;
7. Ganymede [13].

The simplest magnetic-field model is a tilted dipole model of the magnetic field.

$$g_{10} = g_{10}(0) + \frac{dg_{10}}{dt}(t - t(0)) \tag{7.15}$$

$$g_{11} = g_{11}(0) + \frac{dg_{11}}{dt}(t - t(0)) \tag{7.16}$$

$$h_{11} = h_{11}(0) + \frac{dh_{11}}{dt}(t - t(0)) \tag{7.17}$$

$$h_0 = \sqrt{h_{11}^2 + g_{11}^2 + g_{10}^2} \tag{7.18}$$

$$\theta_M = \cos^{-1}\frac{g_{10}}{h_0} \tag{7.19}$$

$$\phi_M = \tan^{-1}\frac{h_{11}}{g_{11}} \tag{7.20}$$

$$b = \frac{h_0 a^3}{r^3}\begin{bmatrix} \sin\theta_M \cos\phi_M \\ \sin\theta_M \sin\phi_M \\ \cos\theta_M \end{bmatrix} \tag{7.21}$$

where t is time, a = 6371 km, and r is the spacecraft distance from the center of the Earth. This model has a linear variation of coefficients with time. Most tables of magnetic fields include the time coefficients.

The Earth's dipole is shown in Fig. 7.6 in a 55-degree inclined orbit in the ECI frame.

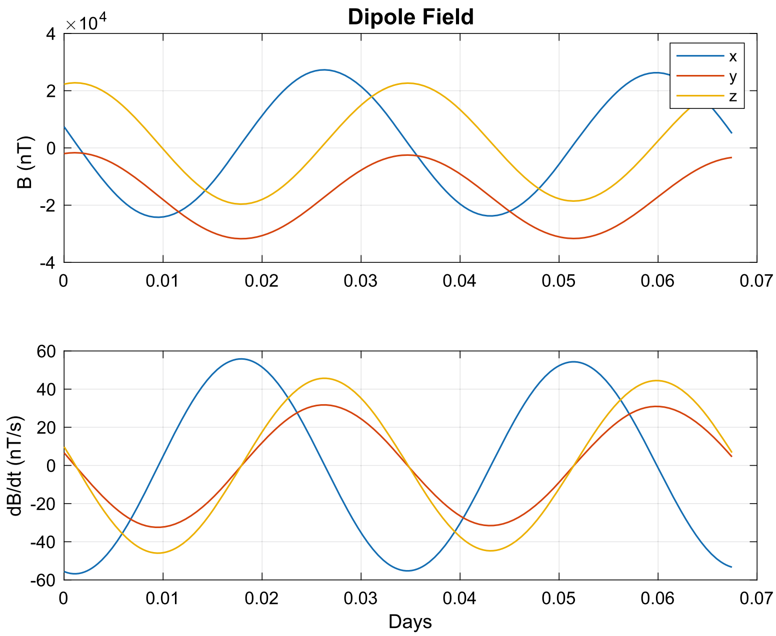
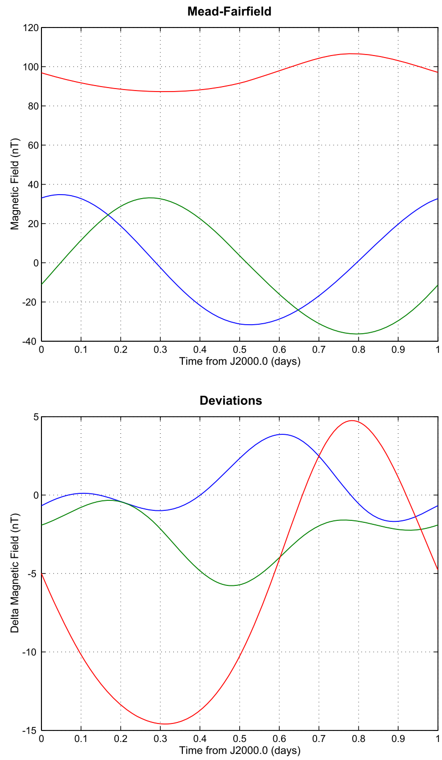


Figure 7.6 The Earth's dipole field in a 55-degree inclined orbit.

The second, suitable for orbits greater than 10 000 km, includes the effect of the Sun and is known as the Mead–Fairfield model [12]. Mead–Fairfield is a quantitative model, based on spacecraft magnetometer data, that perturbs a dipole model to account for the Sun's activity. An input is the solar-activity index. The third is a complete spherical-harmonic expansion of the Earth's magnetic field. Example 7.3 shows the difference between the dipole and Meade–Fairfield model. The Mead–Fairfield model should be used for geosynchronous spacecraft.

```
a  = linspace(0,2*pi);
r  = 42167*[cos(a);sin(a);zeros(1,100)];
jD = JD2000 + linspace(0,1);
BMF( r, jD );
```



Example 7.3: Difference between a dipole and Meade–Fairfield magnetic-field model.

The spherical-harmonic expansion is similar to that of gravity models.

7.8. Ionizing radiation

Ionizing radiation can either be cosmic background radiation or ionizing radiation from radiation belts around a planet. The following planets have radiation belts.

1. Earth;
2. Jupiter;
3. Saturn;
4. Uranus;
5. Neptune.

Jupiter is by far the most intense. The others are on the order of the Earth's [13]. Fig. 7.7 shows the Earth's Van Allen belts.

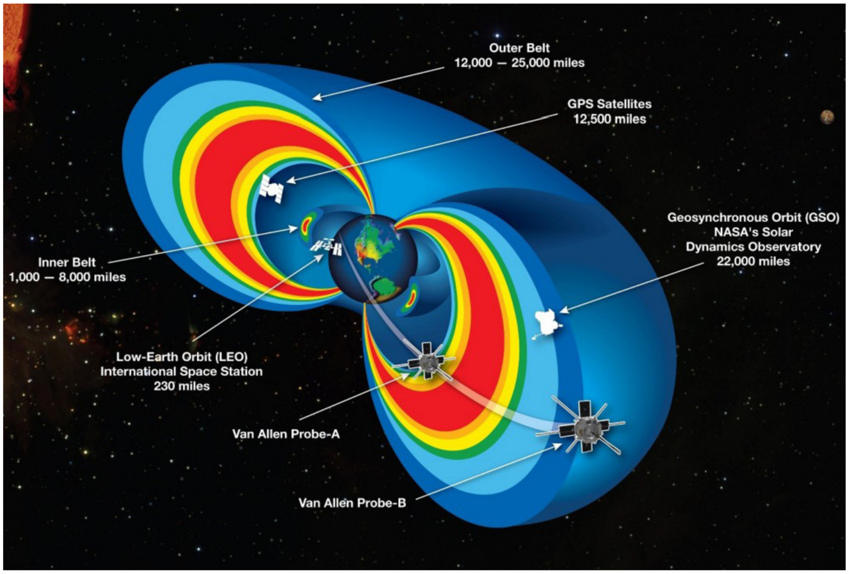


Figure 7.7 The Earth's Van Allen radiation belts. Image Source: http://www.nasa.gov/mission_pages/rbsp/multimedia/20130228_briefing_materials.html.

Fig. 7.8 shows another view with a spacecraft orbit.

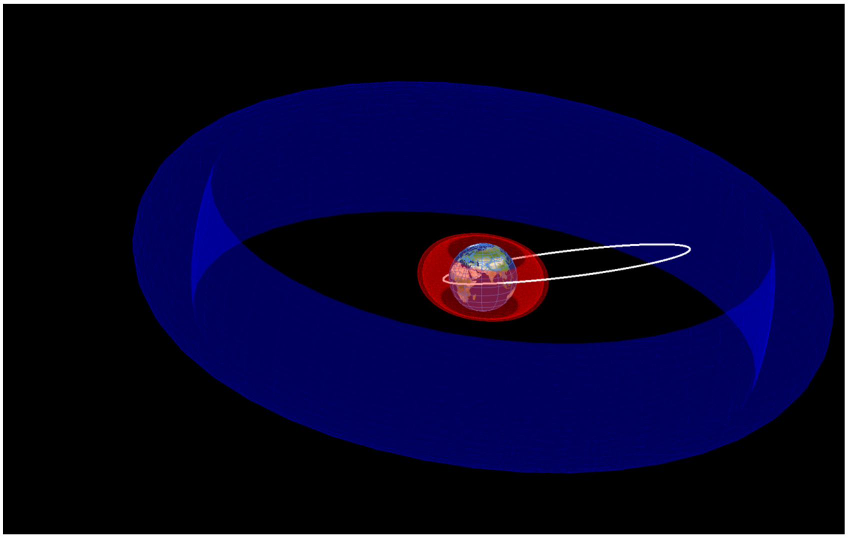


Figure 7.8 Another view showing the inner and outer belts. An inclined orbit is shown for reference.

Ionizing-radiation damage can cause long-term damage to electronics, including solar cells. Ionizing radiation can also have immediate effects:

1. Single-event upsets - when a high-energy particle changes the state of a transistor gate. This is a soft, transient error. It can cause noise in CMOS or CCD chips. A high-energy particle can produce a nuclear reaction at a site. A gate can be destroyed by a high-energy particle.
2. Latch-up - this is a subset of single-event upsets when a particle causes a high current in a transistor that locks up the electronics. This requires cycling the power to clear the problem.

Electronics for attitude control systems should, at the least, be latch-up immune. SEUs for memory are handled by using additional bits to verify the words and by locating the bits on different areas of the memory chip.

Another effect is the degradation of solar cells. The end-of-life power will be lower than the beginning-of-life power. The ACS must be able to operate with lower power levels when the spacecraft is ending its mission life.

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